pi

Fájl: pi.kicad_sch

audio

Fájl: audio.kicad_sch

power

Fájl: power.kicad_sch

battery

Fájl: battery.kicad_sch

ui

Fájl: ui.kicad_sch

Z2Labs – Bassberry (open hardware, CERN-OHL-S-2.0)

Sheet: /

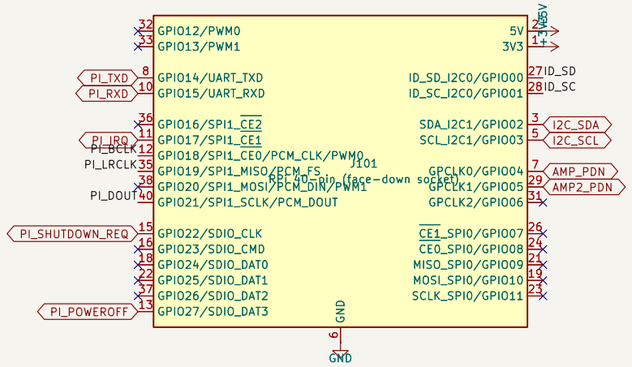
File: bassberry.kicad_sch

Title: Bassberry 2.1 – top level

Size: A3	Date: 2026-10-26	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 1/6

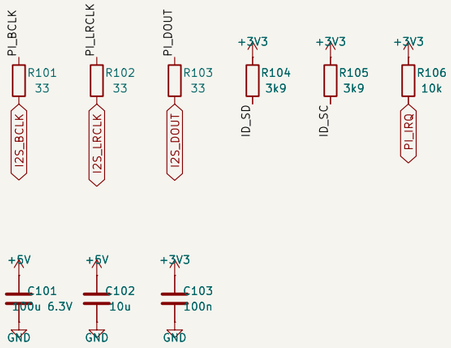
Raspberry Pi plugs in FACE-DOWN: footprint is the mirrored face-down socket.

I2C1 has 1k8 pull-ups on the Pi itself. Series resistors = source termination for I2S.

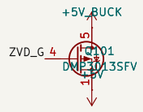


HAT+ ID EEPROM @0x50 (CAT24C32). Write: bridge JP1, then eepflash.sh -w -t=24c32 -a=50 -f=bassberry.eep





Back-power ideal diode (HAT design guide): +5V_BUCK -> +5V (Pi). 5 A rated: DMP3013SFV instead of DMG2305UX.



Z2Labs — Bassberry (open hardware, CERN-OHL-S-2.0)

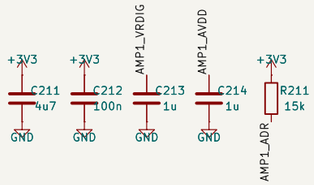
Sheet: /pi/
File: pi.kicad_sch

Title: Raspberry Pi (mounted face-down) + HAT ID

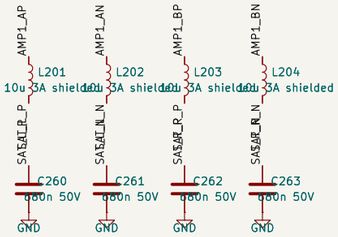
Size: A3	Date: 2026-10-26	Rev: 0.1
KiCad E.D.A. 10.0.6		Id: 2/6

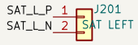
U2 sub amp: PBTL. OUT_A+/OUT_A− and OUT_B+/OUT_B− paralleled pre-filter (verify vs TI datasheet PBTL section).

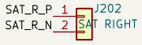
ADR pull-up to DVDD sets the I2C address (TAS5805M datasheet table 7-5): U1 15k -> 0x2D, U2 47k -> 0x2E.

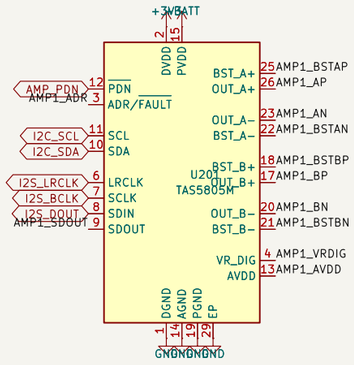


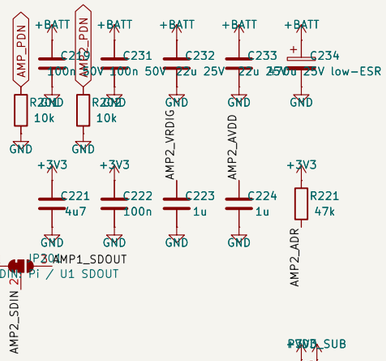


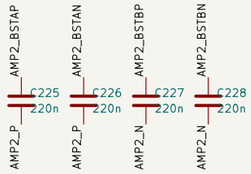


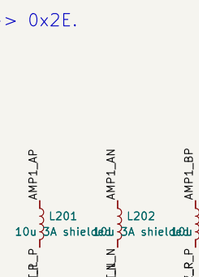


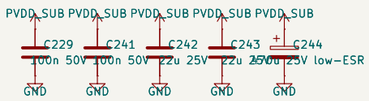










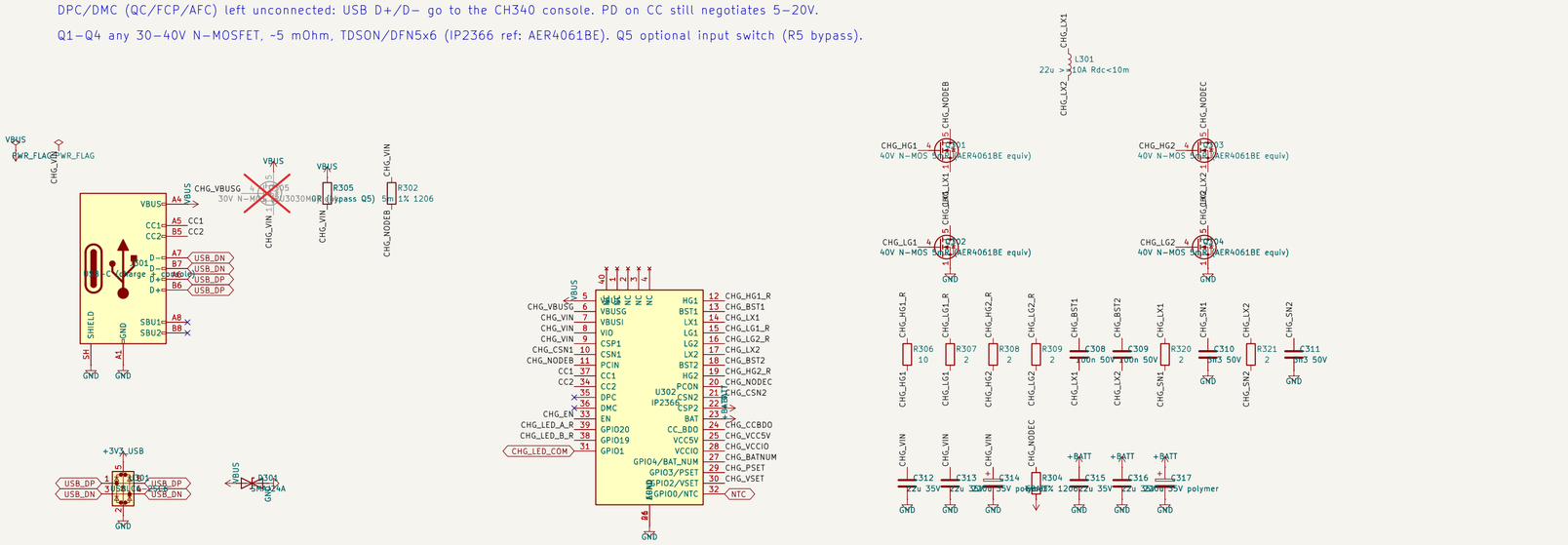


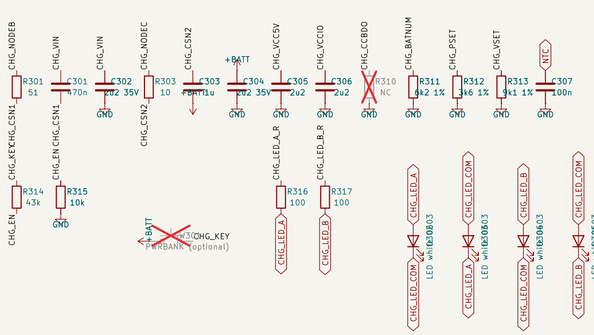
Z2Labs — Bassberry (open hardware, CERN-OHL-S-2.0)	
Sheet: /audio/	
File: audio.kicad_sch	
Title: Audio: 2x TAS5805M (stereo BTL + sub PBTL)	
Size: A3	Date: 2026-10-26
KiCad E.D.A. 10.0.6	Rev: 0.1
	Id: 3/6

IP2366 application per Injoinic datasheet V1.10 fig. 9. Settings: 3S (6.2k), 4.2V/cell (9.1k), 30W (3.6k).

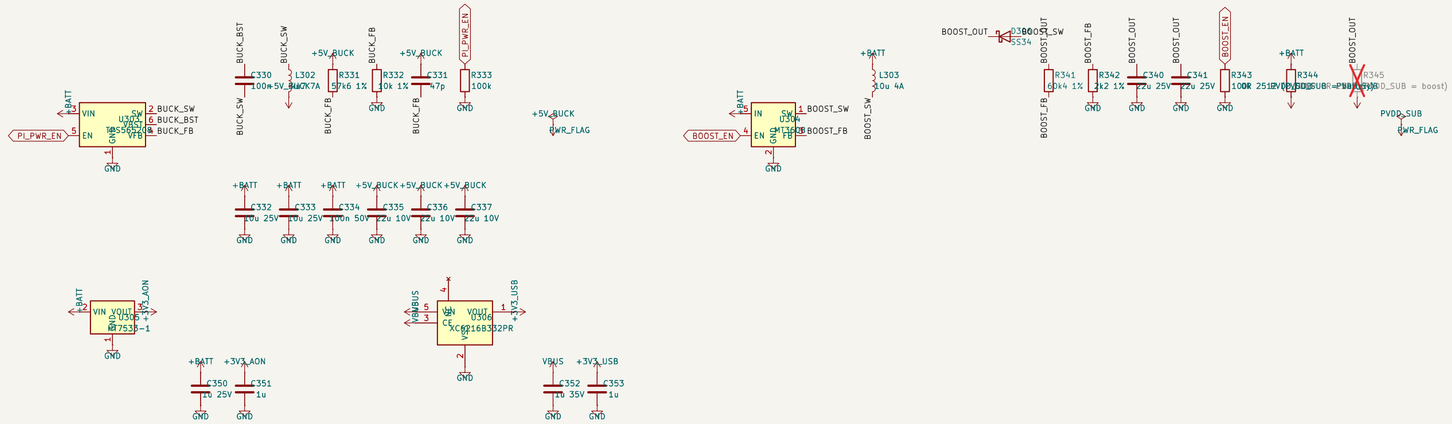
DPC/DMC (QC/FCP/AFC) left unconnected: USB D+/D− go to the CH340 console. PD on CC still negotiates 5–20V.

Q1–Q4 any 30–40V N-MOSFET, ~5 mOhm, TDSON/DFN5x6 (IP2366 ref: AER4061BE). Q5 optional input switch (R5 bypass).





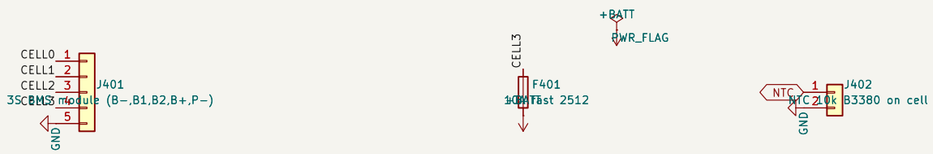
Pi 5V buck (+5V_BUCK, feeds the Pi through the back-power ideal diode on the Pi sheet): Vout = 0.76*(1+R331/R332) = 5.14 V. PI_PWR_EN = VBUS_OK, so the buck only runs when USB-C power is present. R44 (fitted) feeds PVDD_SUB from +BATT, R45 instead from the boost.



3x 18650 in Keystone 1042 snap-in holders (spring contacts; pad 1 = +). Balancing + protection: off-the-shelf 3S BMS module (B-, B1, B2, B+, P-).

Reverse-cell LED lights only when a cell is inserted backwards. P- of the BMS = board GND.

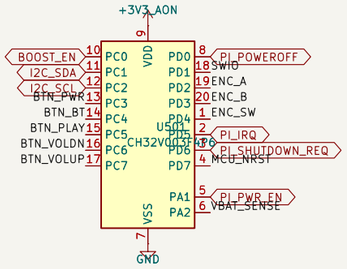


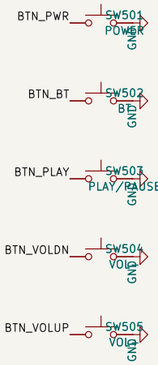


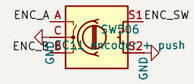
Z2Labs — Bassberry (open hardware, CERN-OHL-S-2.0)		
Sheet: /battery/ File: battery.kicad_sch		
Title: Battery: 3S 18650 clips, BMS module, protection		
Size: A3	Date: 2026-10-26	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 5/6

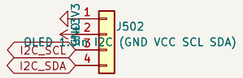
Supervisor runs from +3V3_AON. I2C slave to the Pi on PC1/PC2; PI_IRQ and PI_SHUTDOWN_REQ are open-drain.

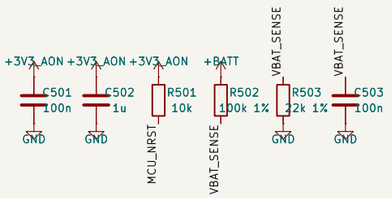
Buttons active-low with MCU internal pull-ups. VBAT_SENSE = VBAT * 22/122.

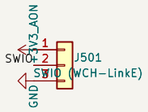


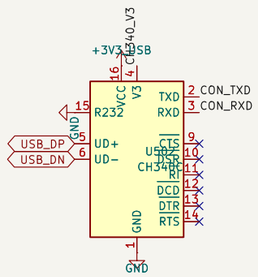


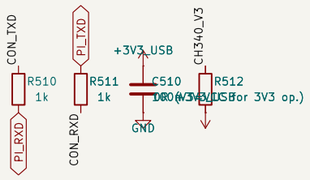












Z2Labs — Bassberry (open hardware, CERN-OHL-S-2.0)

Sheet: /ui/
File: ui.kicad_sch

Title: UI + supervisor MCU (CH32V003), OLED, buttons, encoder, USB console

Size: A3	Date: 2026-10-26	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 6/6